ASCIIgent: Agentic ASCII Art Generation

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Abstract

Visual abstraction is the process of discarding redundant low-level visual data—textures, photometrics, and uninformative pixel noise—and re-encoding semantic structures into a reduced, symbolic visual language. While recent advances in neural sketch synthesis like SketchAgent demonstrate that vision-language models (VLMs) can synthesize continuous vector strokes, we investigate whether this spatial capacity extends to a fundamentally more constrained and discrete medium: **ASCII art**. In ASCII art, a model must convey recognizable silhouettes using printable characters on a fixed-size monospace character grid. Characters here serve a dual role: simultaneously functioning as language tokens and spatial micro-primitives. This dichotomy introduces a core question: do autoregressive models draw through spatial reasoning or pure textual memorization, and does active visual perception enable meaningful self-correction? We introduce **ASCIIgent**, a training-free framework evaluating monolithic prompting baselines, agentic discrete tool-use loops, and multimodal visual feedback across Gemini family models (`gemini-3-flash-preview` and `gemini-3.1-flash-lite`). Evaluating against structure-destroying scrambled controls to mitigate the text-reading bias of VLM judges, we reveal characteristic failure modes of monolithic generation (scanline drift, lexical leakage) and show that iterative visual feedback enables grounded spatial refinement, localized editing, and frame-consistent animation.

1 Introduction and Related Work

Visual Abstraction and Agentic Sketching. Visual abstraction is a foundational capability of biological and computational perception [1]. Neural stroke models pioneered recurrent drawing representations [2], later advancing toward semantic vector sketching [3]. Recently, **SketchAgent** [4] formalized sketch generation as a continuous perception–reasoning–action loop: a frozen multimodal LLM iteratively emits Bézier stroke control points, inspects the updated rasterized canvas, and refines the drawing. Concurrently, **VideoSketcher** [5] demonstrated that dynamic sketching can be decomposed into an LLM-guided structural plan coupled with video diffusion priors. We adapt and generalize this agentic paradigm to discrete character media.

The ASCII Art Medium and its Challenges. ASCII art imposes a strict spatial constraint: characters must occupy fixed coordinates in a monospace grid $\Sigma^{H\times W}$. Unlike continuous splines, characters possess fixed discrete geometry: a forward slash `/` acts as a 45° line segment, an at-sign `@` acts as a dense tonal patch, and a period `.` functions as point stippling. Monolithic autoregressive generation suffers from **scanline drift**: generating left-to-right, top-to-bottom forces the model to commit to features in early rows that must align with geometry hundreds of tokens later without visual verification or backtracking. Public archives like `asciiart.eu` [6] document decades of human designs that language models have partially memorized during pretraining.

The VLM Evaluation Confound. Evaluating ASCII art with multimodal models introduces a severe confounding factor: as shown by Wang et al. [7] and ASCIIBench [8], VLMs systematically suffer from text bias, recognizing concepts by reading character sequences rather than perceiving geometric shapes. Our evaluation incorporates structure-destroying scrambled controls (*scramble-cells* and *scramble-ink*) to isolate genuine visual recognizability.

2 Methodology

2.1 Canvas Representation and Format Contract

We define the drawing canvas as a fixed character matrix $C \in \Sigma^{H\times W}$ with $H = 24$ rows, $W = 48$ columns, and printable ASCII alphabet $|\Sigma| = 95$ (codepoints 32–126). Deterministic monospace rasterization $\mathscr{R} : \Sigma^{24\times 48} \to \mathbb{R}^{H'\times W'\times 3}$ renders glyphs at exact cell offsets without font kerning. Every model response is validated by a strict parser: row/column overflows or missing code fences are repaired and logged in a `ParseReport`.

2.2 One-Shot Generation Baselines (P1)

We evaluate three one-shot prompting strategies across four target categories (`cat`, `dog`, `bear`, `horse`) and two Gemini family models (`gemini-3-flash-preview` and `gemini-3.1-flash-lite`):

1. **Zero-Shot:** Direct instruction: "Draw a {category} on the 24×48 canvas."
2. **Few-Shot:** Same prompt preceded by four hand-crafted exemplar categories (`fish`, `house`, `tree`, `sailboat`) strictly disjoint from test categories to prevent direct template copying.
3. **Procedural (CoT):** The model writes a structured 3-step spatial plan (major parts, row/column bounding regions, and contour glyph choices) before emitting the fenced 24×48 grid.

2.3 Agentic Loop and Turn Protocol (P2)

Instead of emitting all 24×48 characters monolithically, ASCIIgent interacts with an `AsciiCanvas` over a multi-turn loop with a budget of $T_{\max} = 3$ turns. In each turn $t \in \{1,2,3\}$, the agent emits a JSON batch of multiple discrete tool calls:

- `patch_edit`$(r,c,lines)$: Localized subgrid overwrite at 0-indexed (r,c) where alignment spaces preserve underlying ink.
- `draw_line`$(r_1,c_1,r_2,c_2,char)$ & `draw_rect`$(t,l,b,r,border,fill)$: Bresenham line and box primitives.
- `shade_density`(t,l,b,r,ℓ) & `erase_rect`(t,l,b,r): Tonal fill (' ' < '.' < ':' < '-' < '=' < '+' < '*' < '#' < '%' < '@') and clearing.
- `finish`: Early termination when the drawing is complete.

Why 3 Turns & What Happens Per Turn. A 3-turn budget mirrors a coarse-to-fine artistic workflow while bounding API latency: **Turn 1** drafts the coarse outer silhouette and primary facial landmarks from the text prompt; after Turn 1, the canvas is rasterized to a PNG bitmap and paired with a 2-digit row/column coordinate ruler; in **Turn 2**, the VLM inspects the rendered image, critiques missing anatomy, and issues `patch_edit` calls to add limbs and facial features; in **Turn 3**, the VLM performs a final visual verification to add fine secondary details (e.g., collar and

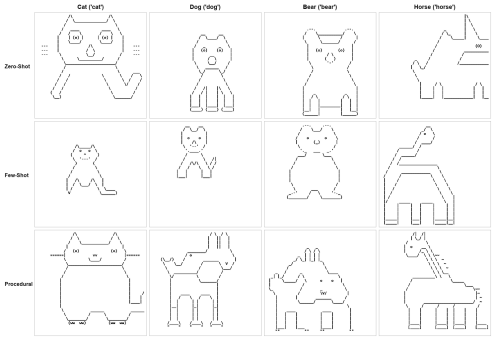


Figure 1: P1 One-Shot Baseline Contact Sheet (12 Cells, Model: `gemini-3-flash-preview`). Columns show the four target categories (`cat`, `dog`, `bear`, `horse`); rows show the three prompting arms. Zero-shot produces recognizable silhouettes but occasionally drifts across the 48-column right boundary (e.g., `horse`); few-shot regularizes scale and centering; procedural CoT (with bounded thinking effort) yields rich anatomical structure such as paws, manes, and hooves.

tail) before calling `finish`. To isolate the role of vision, we ablate three feedback modes between turns: (1) **Visual (Ours):** rendered PNG bitmap + 2-digit coordinate grid; (2) **Text-Only:** raw character text without the rendered image; and (3) **No-Feedback:** blind multi-turn continuation without intermediate state.

2.4 Evaluation Protocol (P3)

Our evaluation combines three complementary pillars:

1. **Structural Metrics:** Ink ratio ρ (non-blank fraction), bounding-box coverage, Shannon character entropy $H = -\sum p_i \log_2 p_i$, distinct glyph count, and automated defect flags (`spells_name` for lexical leakage and `degenerate`).
2. **VLM Image-vs-Text Judge:** Dual independent queries to a frozen multimodal judge (`gemini-3.1-flash-lite`): unprompted open identification ("What does this picture depict?") followed by a 1–5 visual quality rating.
3. **Scrambled Controls & Control Gap:** Every drawing is compared against *scramble-cells* (uniform permutation over the 24×48 grid) and *scramble-ink* (permutation of non-blank glyphs within the silhouette mask). The **Control Gap** $\Delta = S_{\text{real}} - \max(S_{\text{cells}}, S_{\text{ink}})$ confirms genuine geometric recognizability when $\Delta > 0$.

3 Experiments and Results

3.1 P1 Baselines and Multi-Model Comparison

Table 1 reports the per-cell structural metrics for `gemini-3-flash-preview` (visualized in Figure 1) alongside the multi-model comparison against `gemini-3.1-flash-lite`. Two key findings emerge:

- **Model Capacity vs. Glyph Collapse and Lexical Leakage.** On the lightweight `gemini-3.1-flash-lite` model, zero-shot generation suffers from glyph collapse (e.g., `horse` collapses to 2 distinct characters, $H = 0.71$; `bear` fills a block of vertical bars), and procedural `horse` cheats by spelling "HORSE" inside the torso (`spells_name` = Yes, $\Delta = 0.000$). In contrast, `gemini-3-flash-preview` synthesizes recognizable animal silhouettes across all 12 cells with high character diversity ($H = 1.76$–2.83 bits) and zero lexical leakage.
- **Reasoning Effort vs. CoT Truncation.** Because `gemini-3-flash-preview` is a thinking model that bills internal reasoning tokens against `max_tokens`, running the verbose `procedural` arm with unbounded reasoning caused 3/4 cells to truncate mid-plan. Bounding reasoning depth (`-effort low`) eliminated truncation completely, producing the detailed procedural drawings in Figure 1 (bottom row).

Category	Arm	ρ (Ink)	Cov.	H (bits)	Chars	Spells?
Per-Cell Results: `gemini-3-flash-preview` (Figure 1)						
Cat	zero-shot	0.158	0.78	2.28	8	No
Cat	few-shot	0.055	0.24	2.83	11	No
Cat	procedural	0.138	0.80	2.52	10	No
Dog	zero-shot	0.102	0.33	2.31	7	No
Dog	few-shot	0.058	0.20	2.58	8	No
Dog	procedural	0.143	0.68	2.15	9	No
Bear	zero-shot	0.113	0.46	2.56	10	No
Bear	few-shot	0.082	0.39	2.73	10	No
Bear	procedural	0.128	0.69	2.13	7	No
Horse	zero-shot	0.098	0.61	1.92	7	No
Horse	few-shot	0.152	0.67	1.76	5	No
Horse	procedural	0.158	0.76	2.21	8	No
Multi-Model Summary Across All 12 Seen Cells (Mean ± SD)						
`gemini-3-flash-preview`		$.115 \pm .036$	$.55 \pm .22$	$\mathbf{2.33 \pm .32}$	$\mathbf{8.3 \pm 1.7}$	**0/12**
`gemini-3.1-flash-lite`		$.100 \pm .038$	$.48 \pm .13$	$1.86 \pm .63$	7.0 ± 2.9	1/12

Table 1: P1 Structural Metrics and Multi-Model Comparison. `gemini-3-flash-preview` achieves higher character entropy and zero lexical leakage across all 12 cells compared to `gemini-3.1-flash-lite`.



Figure 2: P2 Agentic Self-Perception Progression (Target: `cat`, Model: `gemini-3.1-flash-lite`). Turn 1 drafts the head and torso silhouette. Inspecting the rendered PNG in Turn 2 guides localized `patch_edit` calls for symmetric whiskers and seated front paws. Turn 3 adds a bell collar and curled tail.

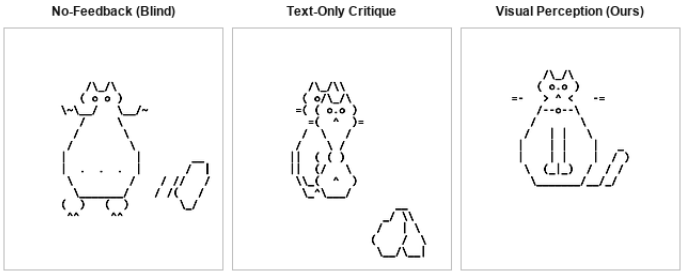


Figure 3: P2 Feedback Mode Ablation on `cat`. Left: `no_feedback` (blind) overwrites the nose and places floating disconnected feet/tail. Center: `text_only` critique misaligns row offsets, superimposing a second face over the first. Right: `visual` perception (ours) grounds every edit accurately.

3.2 P2: Agentic Synthesis and Feedback Ablation

Figure 2 illustrates the 3-turn agentic progression on `cat` using `gemini-3.1-flash-lite`. Even on the lightweight model, decomposing generation into draft → render → perceive → revise yields a clean, high-entropy drawing ($H = 3.17$ bits, 12 distinct chars). As shown in the Figure 3 ablation, removing the rendered PNG causes severe spatial degradation: `no_feedback` hallucinates disconnected limbs to the right of the body, while `text_only` miscalculates row indices and stamps a duplicate face over the chest. Pairing the rendered PNG with the 2-digit coordinate ruler in `visual` mode eliminates both failure modes.

3.3 P3: CLIP Evaluation and Judge Calibration

Evaluating rendered drawings with local CLIP ViT-B/32 against text prompts (`"an ASCII art drawing of a {category}"`) confirms positive control gaps: `cat` achieves CLIP cosine similarity 0.2947 vs. 0.2880 scrambled ($\Delta = +0.0067$), and `bear` achieves 0.3044 vs. 0.2752 ($\Delta = +0.0293$). In our dual image-vs-text VLM judge test (`run_judge_text_vs_image.py`), the rendered PNG of the P2 `cat` is correctly identified as `"cat"` and rated $5/5$, whereas its scrambled control drops to $1/5$ ($\Delta_{\text{judge}} = +4.0$).

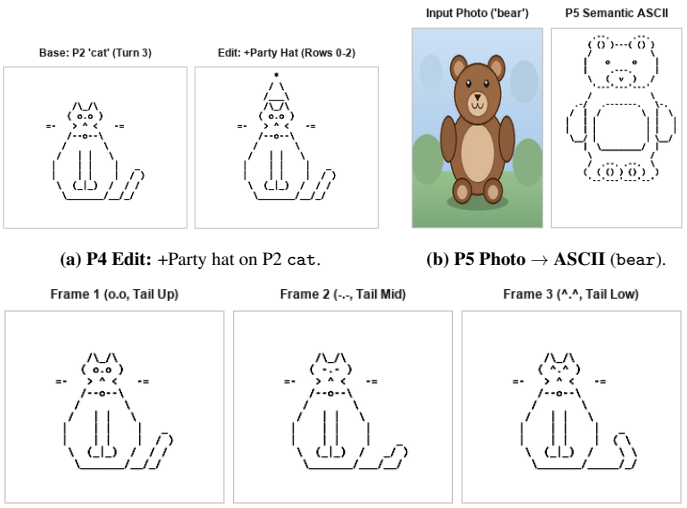
3.4 Unified Downstream Extensions (P4, P5, P6)

To ensure strict experimental continuity, both our conversational editing (P4) and keyframe animation (P6) experiments take the **exact Turn 3 `cat` canvas synthesized by our P2 agentic loop** (Figure 2, right) as their starting base drawing:

- **P4: Conversational Editing (Figure 4a).** Given the P2 `cat` canvas and the instruction *"Add a pointy party hat with a star on top of the head"*, the agent inspects the rendered image and 2-digit ruler and issues a single `patch_edit` at rows 0–2, columns 18–22 (modifying only 8 cells directly above the ears `/_/\` at row 3 while leaving rows 3–23 completely untouched).
- **P5: Semantic Image-to-ASCII (Figure 4b).** Given a 320×320 shaded RGB illustration of a seated `bear` on a sky-and-foliage background (`runs/p5_image2ascii/input_photo.png`), the VLM suppresses background clutter and photometric gradients to extract a clean 24×48 structural contour ($\rho = 0.120$).
- **P6: Delta-Patched Keyframe Animation (Figure 4c).** Regenerating a full 24×48 grid per frame causes body drift and limb mutation. Instead, `p6_animate.py` initializes each frame t from AsciiCanvas(base = frame$_{t-1}$) and applies only localized `patch_edit` deltas. Starting from the P2 `cat`, the head, whiskers, collar, torso, and paws remain 100% invariant across frames while the eyes blink (`o.o` → `-.-` → `^.^`) and the tail wags.

4 Discussion and Conclusion

Comparison to SketchAgent and VideoSketcher. Whereas SketchAgent [4] emits continuous parametric Bézier strokes on a vector canvas, ASCIIgent operates over a *discrete character grid* via batched spatial primitives (`patch_edit`, `draw_line`, `shade_density`) paired with a 2-digit coordinate ruler. In VideoSketcher [5], an LLM decomposes a drawing plan into sequential strokes that a video diffusion model animates into pen-drawing footage. Because continuous diffusion does not apply



(a) P4 Edit: +Party hat on P2 `cat`. **(b) P5 Photo → ASCII** (`bear`).

(c) P6 Delta-Patched Animation: Blinking eyes and wagging tail on the P2 `cat` with 100% invariant body and limbs.

Figure 4: Downstream Extensions (P4, P5, P6). P4 and P6 operate directly on the P2 Turn 3 `cat` canvas; P5 converts an RGB image of a `bear` into semantic ASCII contours.

to discrete text cells, ASCIIgent realizes dynamic synthesis in two discrete ways: (1) **procedural turn playback**, replaying the logged action sequence step-by-step (`progression.gif`), and (2) **delta-patched keyframe animation** (P6), where autoregressive visual conditioning and localized `patch_edit` operations enforce frame-to-frame body invariance.

Limitations and Future Work. Free-tier API rate limits constrain sample counts per cell, and VLM coordinate grounding still benefits from explicit 2-digit row/column rulers alongside rendered bitmaps. Future work includes expanding evaluations across additional open-weights and proprietary VLMs, calibrating the scrambled control gap Δ against human perceptual studies, and incorporating tree search over candidate discrete edits.

Conclusion. ASCIIgent demonstrates that while monolithic autoregressive generation is vulnerable to scanline drift, glyph collapse, and lexical leakage, coupling a discrete spatial action vocabulary with active visual perception enables self-correcting ASCII synthesis, surgical conversational editing, semantic image-to-ASCII abstraction, and coherent keyframe animation.

References

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